Diagnosing and Mitigating Fitting to Confounding EEG Signals

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***Abstract*—Non-invasive brain-computer interfaces rely on classification models trained on a dearth of high-quality EEG data collected by researchers. However, models trained on data from highly controlled experiments with low ecological validity may fit to features based on stimuli used in the experiment, rather than the intended target behaviours. Models are prone to fitting to non-interest features introduced in the experiment design, such as directional attention, event-related syncronization/desyncrhonization and thus overestimate generalizability. We briefly introduce existing EEG literature on overfitting to non-interest stimuli. Then, we investigate the effect of removing non-interest intervals on a convolutional neural network's feature importance.**

I. Introduction

Electroencephalography (EEG) uses electrodes on or near the scalp to measure near-surface brain activity. Machine analysis of highly dense and complex EEG data has proven to be effective; techniques to classify mental states and behaviours from EEG continue to improve with new research.

Much current research in classification of EEG seeks to classify behaviours through the training of machine learning algorithms. To maximize performance and decrease computational training costs, machine learning algorithms usually train on high quality data, recorded in highly controlled environments. This is appropriate procedure to ensure reproducibility in experiments.

However, it is a well-known pitfall that machine learning algorithms will fit to unintended patterns such as noise, diminishing its generalizability. Specific to EEG, unintended patterns include oculomotor artifacts [1], cardiac signal [2], power line [3], which require expert knowledge in addressing and lacks a general optimal algorithm for all types of artifact [4].

Especially with techniques where interpretability is challenging, such as neural networks, especially shallow and deep neural networks, overfitting to a consistent feature present in all samples is difficult to diagnose. More severely in the scenario of reproducible and consistent EEG experiments, machine learning algorithms will fit to stimuli unique to the experimental design [5], [6], [7]. These include event-related desyncronization/synchronization (ERD/ERS)[1] and visual/ auditory cueing during data collection.

While the relation between muscle and cortical area are more complex than one-to-one, previous research on classification of motor imagery has shown that the most significant locations of EEG signals are the motor cortex areas corresponding to their respective muscle control areas [1].

Without expertise in EEG, researchers may rely on machine learning algorithms to detect and mitigate confounding features automatically, using techniques such as dropout [8] and nearest-neighbors [9].

With expertise in EEG, a researcher aware of this problem may utilize one or more of the following techniques to identify or mitigate the presence and influence of confounding signals, depending on the experiment design:

1) Identify and quantify the effect of the signal corresponding to the experiment cueing. A minimal effect can be neglected [1].
2) Filter with bandpass to remove expected range of confounding component [10]. However, some confounding features cannot be completely excluded by a single frequency band [7].
3) Manually separate the non-interest data from interest data.

Manual intervention is potentially capable of saving much computational cost. We investigate manually separating the data containing the target feature and confounding features to demonstrate the predictiveness of each feature and the effect on the behaviour of a model.

II. Methods

We attempt to classify the left or right motor imagery (MI) from the recorded EEG of participants in controlled studies.

As the goal of this paper is not performance but the manipulation of feature importance, a lightweight and small convolutional network model is trained with 300 epochs. For this purpose, we use a shallow convolutional network [10]. The architecture of this network is designed for performance in motor imagery classification. Superior computational power and training has demonstrated that with a similar model, the expected accuracy on the whole dataset should be 66.79±16.34 [11].

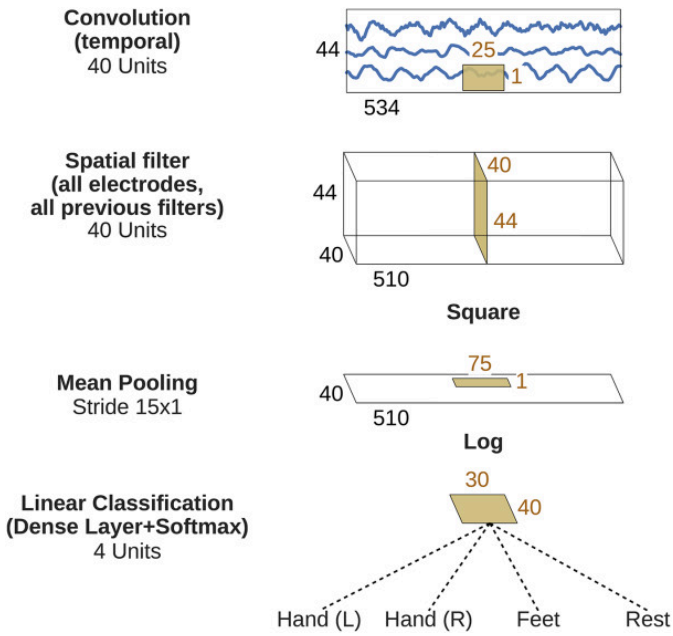

Fig. 1. Architecture of the shallow convolutional network. From Schirrmeister et al. [10]

However, it is not immediately obvious which features are most important in that model. In line with similar literature and manual inspection of the dataset [1], we expect the electrodes at the posterior parietal cortex to be the most significant features in our model. These are electrodes position C3, C4 on the international 10-10 system, but a significance of electrodes at the occipital regions should also be observed, since the cueing stimuli will evoke a response in subjects.

We train two models on one of two subintervals of each trial in the dataset. Both subintervals are the same size and timespan.

a) The first subinterval includes the period 100ms seconds prior to the cue until 400ms seconds after the cue. We attempt to include only the data that is the robust confounding feature. This is the stimuli interval.

b) The second subinterval includes the period 500ms after the cue until the end of the trial, 500ms later. We attempt to include only the data that is the intended feature. This is the motor imagery interval.

We use the same visualization methodology as the original paper which introduced shallow convolutional networks for EEG [10]. Gaussian noise is randomly introduced to perturb inputs, illustrating the importance of spatial features on a topographic map.

Hyperparameters are similar to the values recommended by MOABB, using the Adam optimizer [12]. The learning rate scheduler follows a recursive form of the cosine annealing schedule:

$$\eta_{t+1} = \eta_{\min} + (\eta_t - \eta_{\min}) \cdot \frac{1 + \frac{\cos((T_{\text{cur}}+1)\pi)}{T_{\max}}}{t + \cos\left(\frac{T_{\text{cur}}\pi}{T_{\max}}\right)} \tag{1}$$

where η_t is the learning rate at epoch t, T_{cur} is the number of epochs since the last restart, $T_{\max}$ is the maximum number of epochs in the training process, as implemented in the PyTorch library. Dropout probability is set to 0.6.

III. Dataset

We use openly available recordings from Cho et al. [1]. Datasets are provided by the open-source MOABB project [11]. Both datasets record whether the participant is imagining raising their left or right hand, with different cueing stimuli. In the experimental design described by Cho et al., participants are cued to perform motor imagery on either the left or right based on a displayed verbal cue.

Original trials are recorded at 512 hz. Trials are resampled to 200 hz.

IV. Results

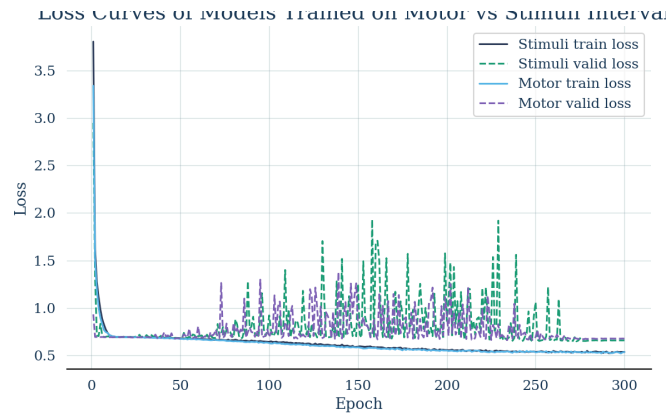

Fig. 2. Loss curves for both stimuli and motor intervals. The validation loss curve of the model trained on the stimuli interval is less stable.

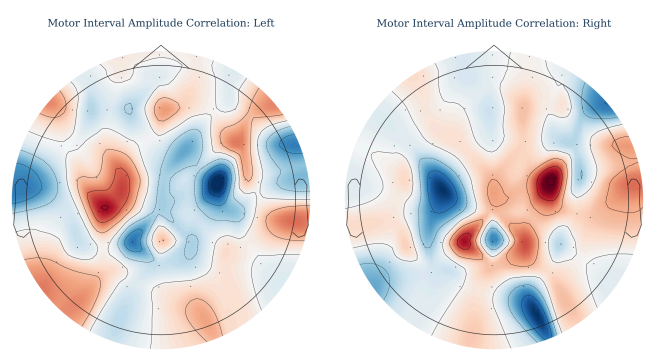

Fig. 3. Spatial significance of amplitude from the shallow convolutional network trained on the motor imagery interval.

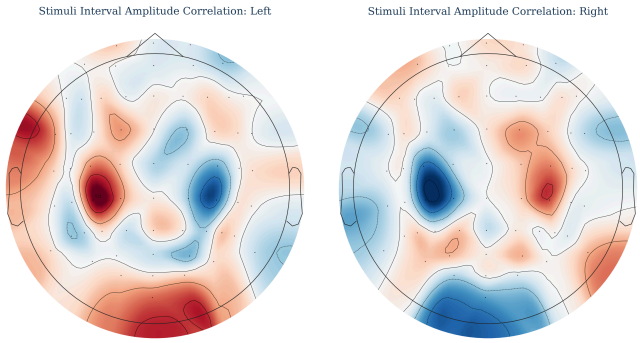

Fig. 4. Spatial significance of amplitude from the shallow convolutional network trained on the stimuli interval.

Both models figure 4 show a strong separation between left and right classes.

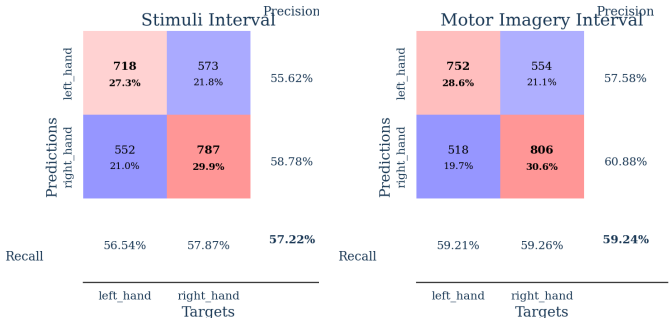

Fig. 5. Confusion matrices for the model trained on the stimuli and motor imagery and interval.

V. Discussion

Our experimental design selected one-sixth of the original trial data. By selecting the interval where we expect our indended target feature to be present, we retain much of the predictiveness while excluding much confounding patterns from our dataset. This had the added benefit of massively diminishing computational cost, allowing training of the model to be performed on a lightweight laptop.

Our results showed a significant separation between the features in left and right classes. The significance of the amplitudes in motor cortical areas were higher in the motor imagery interval than in the stimuli interval.

However, even in the stimuli interval, amplitudes in the posterior parietal motor area are still a significant feature. In addition, the confusion matrices appear very similar. This may suggest that subjects are capable of reacting with motor imagery faster than expected in this experiment.

In the motor imagery interval, amplitudes in a small occipital area (expected for processing visual stimuli) are significant features, albeit less significant than posterior parietal amplitudes. This may suggest that evoked responses to experimental cues are more enduring than expected in this experiment. This is supported by Cho et al., who find that event-related desynchonization effects in the occipital areas persist from 500ms after the cue.

EEG signal has a poor signal-to-noise ratio. The stability of the loss curve is superior in the model trained on the intended feature, however, the validation loss and training loss are nearly identical at the end of training. Despite being ~2.02% more accurate, validation loss is very slightly higher in the model trained on the motor imagery interval. It is possible that the splits were not the best representatives of the dataset.

VI. Future work

For reproducibility, the MOABB project provides benchmarking functions. However, in the leave-one-out evaluation paradigm, training of convolutional neural networks can become very computationally expensive. With more computational power, we could demonstrate that the convolutional neural network could be more consistent in terms of predictiveness in the motor imagery interval vs the stimuli interval.

To tease apart the relationship between the intended and confounding features on, a combination of spatial and temporal perturbation may be more informative than our approach, which illustrated only spatial amplitude significance. One may also attempt to compute a power spectrum decomposition, since motor and stimuli features belong primarily, but not entirely, to separate power bands [1], [7].

Our experiment used the same hyperparameters between intervals. It is possible that different hyperparameters would give different results.

We expect combination of datasets from different experimental designs to diminish the robustness of experimental design effects [7]. However, among available data, experimental designs are very standardized, and code defining experiment paradigms are not always publicly available.

Existing datasets with different cues include Zhou et al. [13]. As this dataset has one of the highest accuracy among datasets catalogued on MOABB, it may be worth investigating whether the significant features of this dataset are the same as word-cued datasets such as Cho et al., used in this experiment. Prior experiments demonstrate that the processing of direction is faster when using arrow cues than when using word cues [14], [15], [16].

In addition, many existing datasets catalogued by MOABB have superior or equal performance in Riemannian geometry methods [11], [17]. These happen to be much less computationally expensive than deep learning methods.

Our attempt to diminish the robustness of the confounding features was not significantly successful in improving the learning rate. Because responses to stimuli are very heterogeneous, data from a single subject, or subjects that are more similar in their responses, may improve learning rate and predictiveness of a model [1].

As described in our introduction, experiments are highly controlled with the same cueing stimuli across all trials. In line with similar recommendations by Xu et al. [7], we propose that dataset collection experiments utilize a variety of cueing paradigms (auditory, haptic, visual) per participant. We suggest that models trained on poorly controlled, naturalistic data with naturally occuring task cues [18], [19] may be more robust against fitting to confounding features.

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